



WORKFORCE ATTRITION ANALYTICS DASHBOARD

Human Resources Analytics Project

Advanced Workforce Analytics & Attrition Cost Modelling



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Institution:

STARTECHONE NIGERIA LIMITED



June 2026

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Atlas TechWorks - Case Study

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Atlas TechWorks is a 1,500-employee technology company experiencing rising voluntary employee attrition across several departments, job levels, and employee groups. High employee turnover not only disrupts productivity and team stability but also increases recruitment, onboarding, and replacement costs.

This project analyzes workforce data to uncover the key drivers of employee attrition, measure the financial impact of employee departures, and identify the employee segments most at risk of leaving. Using Power BI, the analysis explores demographic patterns, workplace factors, promotion stagnation, and attrition costs to provide actionable recommendations that can improve retention and reduce organizational losses.

By transforming workforce data into meaningful insights, Atlas TechWorks can make informed people-management decisions and prioritize interventions that deliver the highest return on investment.

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Aims

To analyze employee attrition at Atlas TechWorks using Power BI in order to identify attrition patterns, determine key drivers of employee turnover, quantify the financial impact of attrition, and recommend data-driven retention strategies to improve workforce stability and organizational performance.

Objectives

- To determine the overall attrition rate and analyze how attrition varies across departments, job roles, age groups, tenure bands, and job levels.
- To identify the key factors influencing employee attrition, including overtime, job satisfaction, work-life balance, distance from home, and compensation levels.
- To evaluate the relationship between promotion stagnation and employee attrition, particularly among lower job levels.
- To estimate the annual financial cost of attrition using employee salary data and replacement cost assumptions.
- To identify departments and employee groups that contribute most significantly to attrition-related costs.
- To develop interactive Power BI dashboards that provide clear visual insights into attrition trends, drivers, and costs.
- To model potential retention scenarios and estimate cost savings that could be achieved through targeted attrition reduction strategies.
- To provide actionable recommendations that support workforce planning, employee engagement, and long-term retention initiatives.

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HEADCOUNT



1,500

ATTRITION RATE



30.13%

TOTAL ATTRITION



452

Total Attrition Cost



\$363.2M

1. Headcount – Total Active Workforce

Insight: Atlas TechWorks employs 1,500 people across multiple departments and business functions. While this represents a sizeable workforce capable of supporting business growth, maintaining workforce stability is critical. High employee turnover within a workforce of this size can create significant operational disruption, increase recruitment demands, and place additional pressure on remaining employees.

2. Attrition Rate – Percentage of Employees Who Left the Organization

Insight: The organization records an attrition rate of 30.13%, meaning nearly one in every three employees has left the company. This level of turnover is considerably high for a technology-driven organization and signals underlying retention challenges. Sustained attrition at this level can negatively affect productivity, employee morale, organizational knowledge retention, and long-term business performance.

3. Total Attrited Employees – Number of Employees Lost

Insight: A total of 452 employees have left the organization. This represents a substantial loss of talent and organizational experience. Beyond replacement costs, employee departures can create project delays, increase workload pressure on remaining staff, and weaken team continuity across key business functions.

4. Annual Attrition Cost – Estimated Cost of Employee Turnover

Insight: Employee attrition is estimated to cost Atlas TechWorks approximately \$363.2 million annually. This figure reflects the financial burden associated with replacing employees, including recruitment, onboarding, training, productivity loss, and knowledge transfer costs. The scale of this cost demonstrates that attrition is not only a workforce challenge but also a significant financial risk to the organization.

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Avg Attrition Cost per E...

 \$803.6K

% Payroll Lost

 30.15%

Total Annual Payroll

 192M

5. Cost per Employee Lost – Average Cost of Replacing a Departing Employee

Insight: Each employee departure costs the organization an estimated \$803.6 thousand on average. While individual replacement costs vary by job level and role complexity, the cumulative impact becomes substantial when turnover occurs at scale. Reducing even a small percentage of employee departures could generate meaningful cost savings.

6. Payroll Lost – Percentage of Payroll Impacted by Attrition

Insight: Approximately 30.15% of annual payroll value is effectively lost through employee turnover. This indicates that a significant portion of workforce investment fails to generate long-term returns due to employee departures. Lowering attrition would help protect payroll investments while improving workforce productivity and organizational stability.

7. Total Payroll – Annual Employee Compensation

Insight: Atlas TechWorks invests approximately \$192 million annually in employee compensation. Given the high attrition rate observed across the workforce, protecting this investment through improved retention strategies should be a strategic priority. Retaining experienced employees often delivers greater value than repeatedly replacing lost talent.

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POTENTIAL ANNUAL SAVINGS

\$0.0

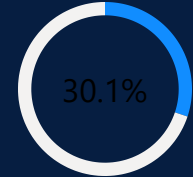
EMPLOYEE RETAINED

0

PROJECTED ATTRITED EMPLOYEES

452

Projected Attrition Rate



1. Potential Annual Savings: Potential Annual Savings represents the estimated reduction in annual attrition costs that could be achieved if the selected intervention successfully lowers employee turnover.

Insight: This KPI helps leadership understand the financial value of retention initiatives before investing resources. Higher savings indicate that the selected attrition driver contributes significantly to employee departures and should therefore be prioritized.

2. Employees Retained: Employees Retained represents the estimated number of employee departures prevented under the selected intervention scenario and implementation intensity.

Insight: This metric translates retention strategies into workforce impact. As intervention effectiveness increases, more employees remain within the organization, reducing recruitment demands, preserving institutional knowledge, and improving workforce continuity.

3. Projected Attrited Employees: Projected Attrited Employees represents the expected number of employees who would still leave the organization after applying the selected retention initiative.

Insight: This KPI provides a realistic estimate of remaining attrition risk. It allows decision-makers to assess whether a single intervention is sufficient or if multiple retention strategies must be combined to achieve meaningful improvements.

4. Projected Attrition Rate: Projected Attrition Rate measures the estimated workforce attrition percentage after applying the selected intervention and implementation intensity.

Insight: This metric serves as the primary success indicator for retention initiatives. A lower projected attrition rate indicates improved workforce stability and reduced employee turnover. Management can use this KPI to compare intervention scenarios and identify the most effective retention strategy.

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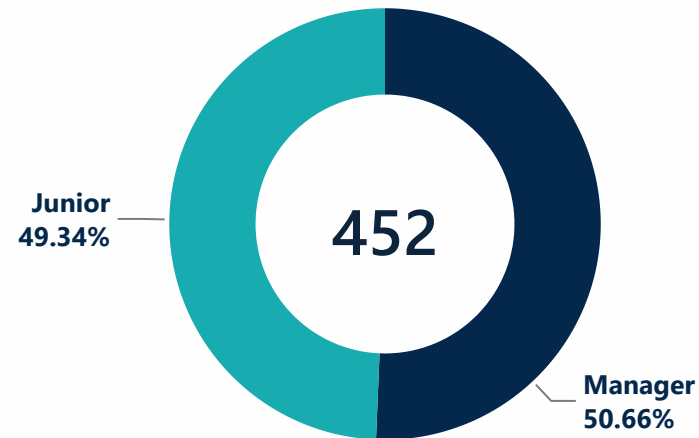


Forecasting

Atlas TechWorks - Mangers and Junior Employees Experience Nearly Equal Levels of Attrition

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Attrited Employees by JobLevel



Insight

Attrition is distributed almost evenly between managerial and junior-level employees, with 50.66% of managers and 49.34% of junior employees leaving the organization. This balanced distribution suggests that employee turnover is not isolated to a specific organizational level but is affecting the workforce broadly.

The slightly higher attrition among managers is particularly concerning because leadership departures often create wider operational disruptions, knowledge gaps, and increased replacement costs. The findings indicate that retention strategies should focus not only on entry-level employees but also on preserving experienced talent in leadership positions.

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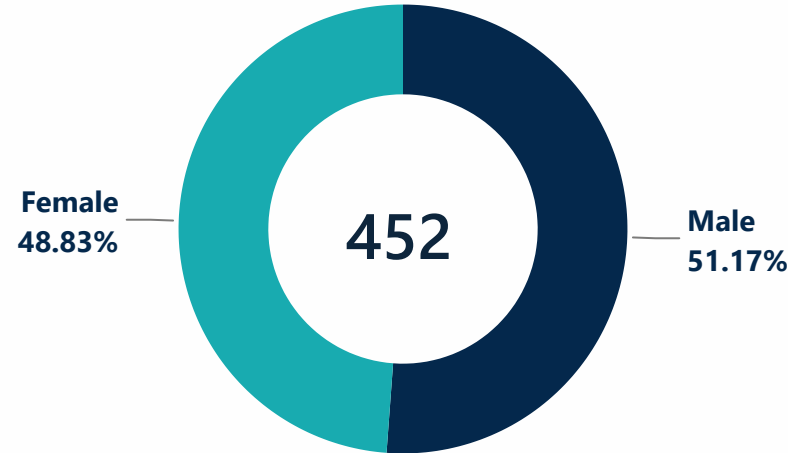


Forecasting

Atlas TechWorks - Male Employees Record Slightly Higher Attrition Than Female Employees

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Attrition Rate by Gender



Insight

Male employees exhibit the highest attrition rate at 51.17%, compared to 48.83% among female employees. While the difference is relatively small, the pattern suggests that workforce turnover is affecting both genders at similar levels.

The narrow gap indicates that attrition is likely being driven more by workplace conditions, career growth opportunities, and employee experience factors rather than gender-specific issues. Organization-wide retention initiatives may therefore deliver greater impact than gender-targeted interventions.

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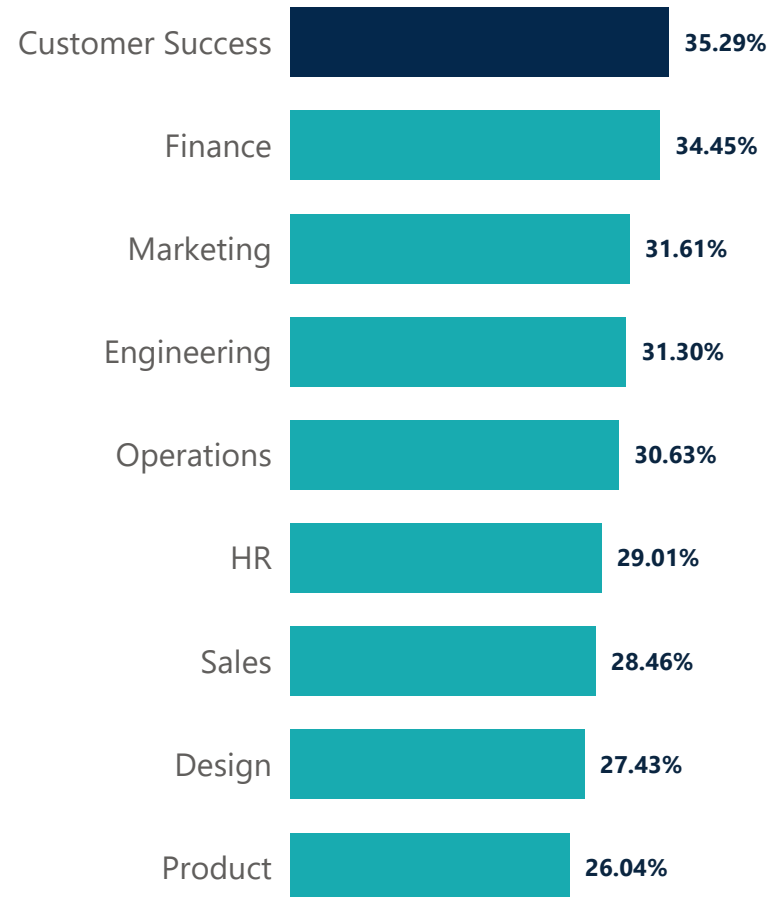


Forecasting

Atlas TechWorks - Customer Success and Finance Experience the Highest Attrition Rates

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Attrition Rate by Department Name



Note: Percentages are calculated independently for each department

Insight

Customer Success records the highest attrition rate at 35.29%, closely followed by Finance at 34.45%. Marketing (31.61%) and Engineering (31.30%) also exceed the organizational average, indicating elevated turnover within several key operational functions.

Product (26.04%) and Design (27.43%) report the lowest attrition rates, suggesting comparatively stronger employee retention within these teams.

The variation across departments highlights potential differences in workload, career progression opportunities, management practices, or employee engagement.

Departments with the highest attrition should be prioritized for further investigation and retention-focused interventions.

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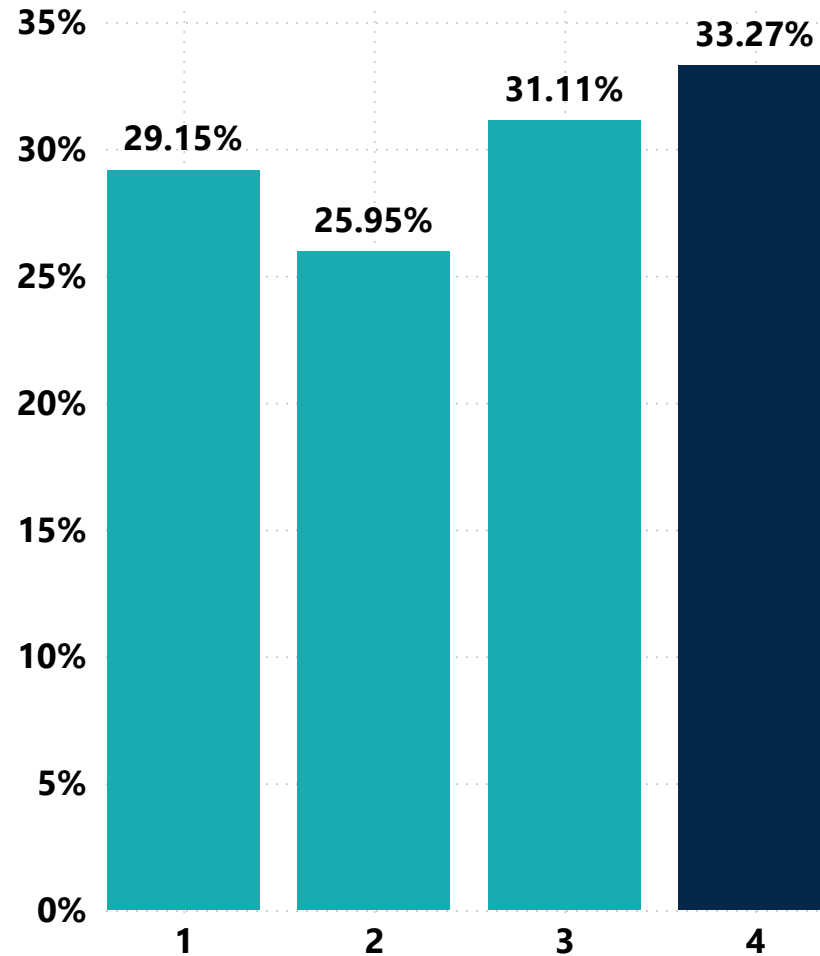


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Atlas TechWorks - Long-Tenured Employees Show the Highest Propensity to Leave

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Attrition Rate by Tenure Band



Insight

Employees with more than ten years (4) of service record the highest attrition rate at 32.27%, followed closely by those with six to ten years (3) of tenure at 31.11%.

In contrast, employees with two to five (2) years of service experience the lowest attrition rate at 25.95%.

This pattern suggests that retention challenges are not limited to new hires. Employees who remain with the organization for extended periods may experience career stagnation, limited advancement opportunities, or declining engagement, increasing their likelihood of departure.

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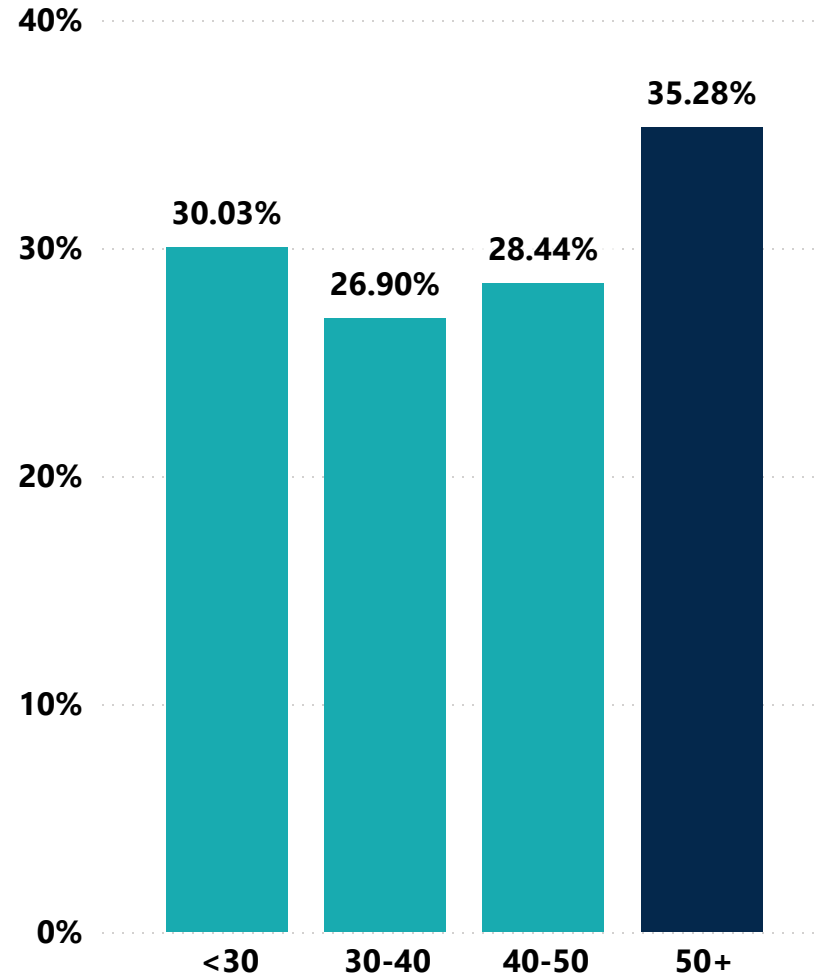


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Atlas TechWorks - Employees Aged 50 and Above Exhibit the Highest Attrition Levels

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Attrition Rate by Age Band



Insight

Employees aged 50 years and above record the highest attrition rate at 35.28%, significantly exceeding all other age bands.

Employees between 30 and 40 years' experience the lowest attrition rate at 26.90%, while employees under 30 report an attrition rate of 30.03%.

The elevated turnover among older employees may indicate retirement transitions, career changes, or unmet expectations regarding advancement and workplace flexibility.

Retention initiatives focused on experienced employees may help preserve institutional knowledge and leadership expertise.

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Atlas TechWorks - Employees in the Support Engineer role within the customer success department record the highest Attrition rate.

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Top Ten Job RoleName by Attrition Rate

JobRoleName	DepartmentName	Attrition Rate	Attrited Employees
Support Engineer	Customer Success	38.98%	23
Software Engineer	Engineering	38.71%	24
Sales Manager	Sales	36.62%	26
Product Manager	Product	36.54%	19
Marketing Specialist	Marketing	36.51%	23
Financial Analyst	Finance	36.36%	20
Design Lead	Design	35.82%	24
Account Executive	Sales	33.33%	19
Controller	Finance	32.81%	21
Senior Engineer	Engineering	32.08%	17

INSIGHTS

The role-level analysis reveals that Support Engineers (38.98%) and Software Engineers (38.71%) experience the highest attrition rates, making them the most vulnerable positions within Atlas TechWorks. Sales Managers, Product Managers, Marketing Specialists, and Financial Analysts also exhibit attrition rates above 36%, indicating that turnover is spread across both technical and business functions.

Although Sales Managers recorded the highest number of exits (26 employees), Software Engineers and Design Leads each lost 24 employees, while Support Engineers and Marketing Specialists recorded 23 departures. These findings suggest that attrition is affecting several critical roles simultaneously rather than being concentrated in a single area.

At the lower end, Controllers (32.81%) and Senior Engineers (32.08%) show relatively lower attrition rates but still experience considerable employee losses. The consistently high turnover across key positions highlights the need for targeted retention strategies, particularly for customer-facing and technical roles where employee replacement costs and productivity disruptions are often significant.

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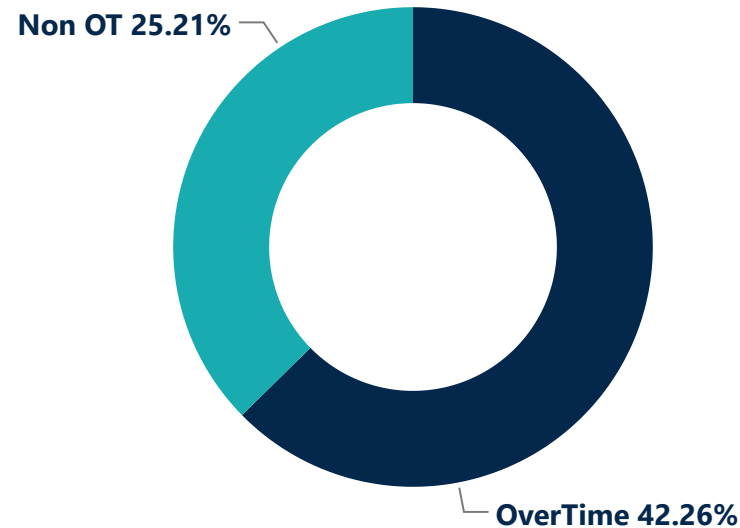
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Attrition Rate by Overtime



Insight

Employees who regularly work overtime experience an attrition rate of 42.26%, compared to just 25.21% among employees who do not work overtime.

This 17 percentage point gap makes overtime one of the strongest predictors of employee turnover within the organization.

The findings suggest that prolonged workloads, increased stress levels, and reduced work-life balance may be contributing significantly to employee departures. Reducing excessive overtime could have a substantial impact on improving retention rates.

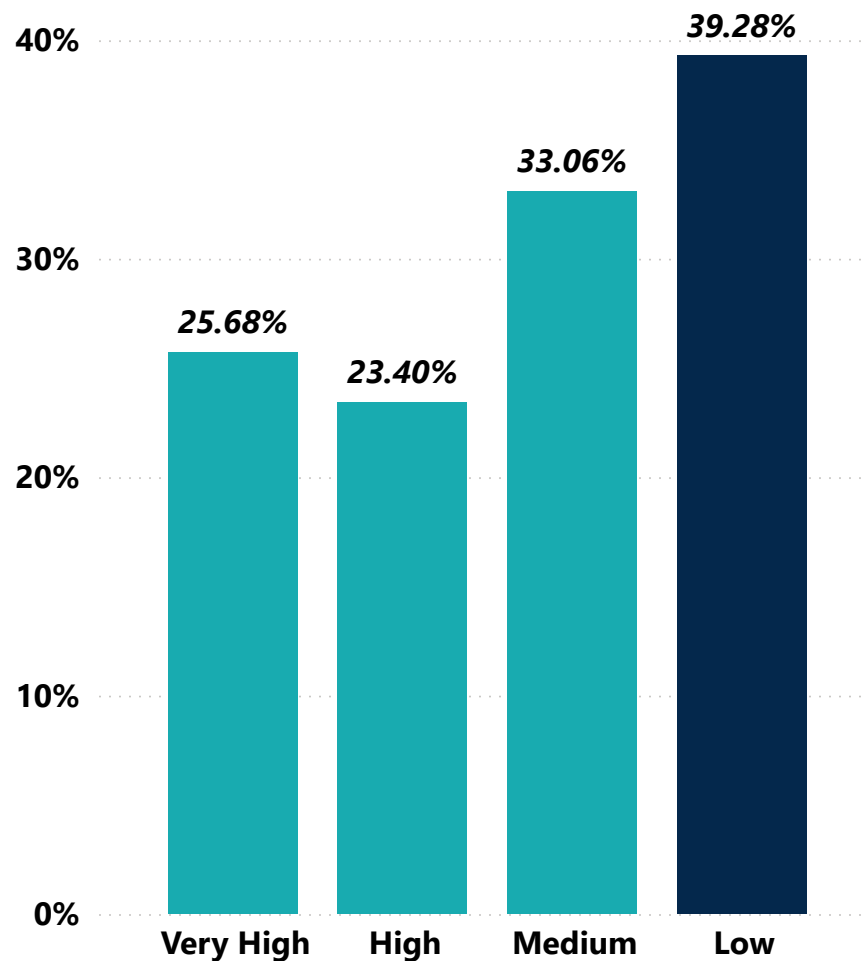
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Atlas TechWorks - Low Job Satisfaction Is Closely Linked to Higher Attrition

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Attrition Rate by Job Satisfaction



Insight

Employees reporting low job satisfaction experience the highest attrition rate at 39.28%. Attrition gradually declines as satisfaction levels improve, with highly satisfied employees recording a significantly lower attrition rate of 23.40%.

The relationship demonstrates a strong connection between employee satisfaction and workforce retention. Employees who feel disengaged, undervalued, or unsupported are considerably more likely to leave the organization.

Improving employee experience and engagement could therefore generate meaningful reductions in attrition.

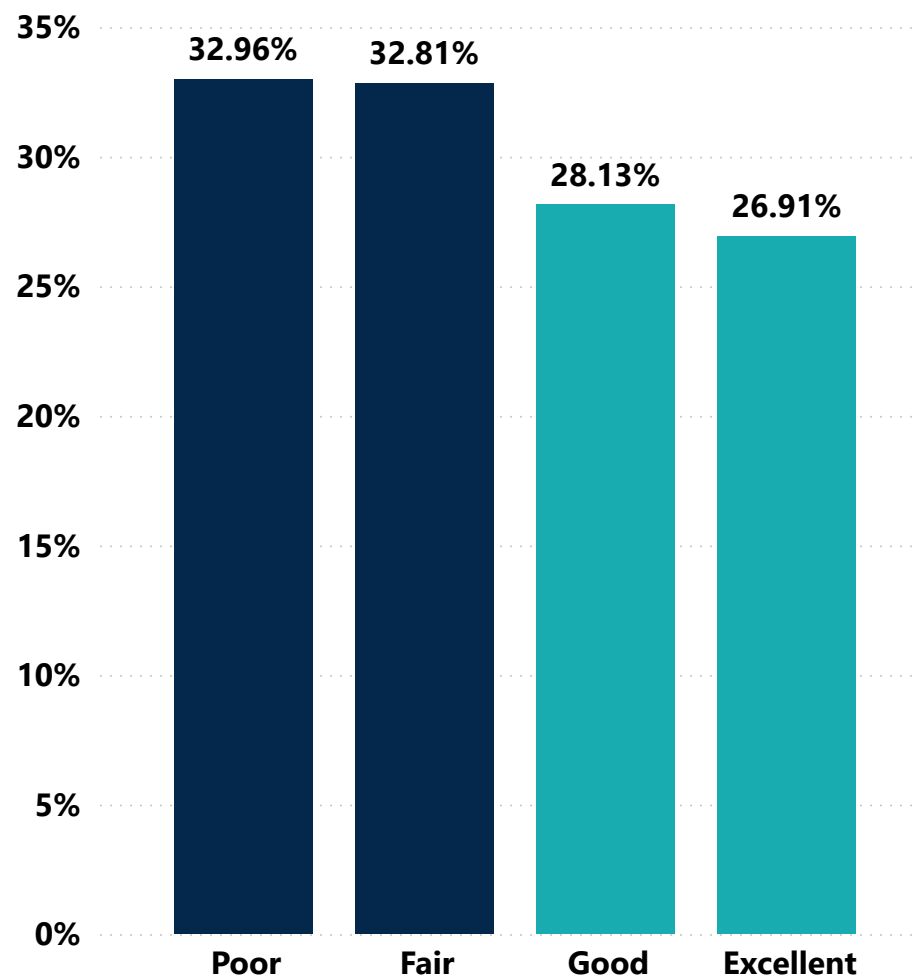
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Atlas TechWorks - Better Work-Life Balance Corresponds with Lower Attrition

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Attrition Rate by WorkLifeBalance



Insight

Employees reporting poor work-life balance experience the highest attrition rate at 32.96%, closely followed by those with fair work-life balance (32.81%) while those reporting excellent work-life balance record the lowest rate at 26.91%.

Although the gap is smaller than that observed for overtime and job satisfaction, the trend remains consistent. Attrition decreases as employees perceive better balance between professional and personal responsibilities.

These findings reinforce the importance of flexible work arrangements, workload management, and employee wellbeing initiatives.

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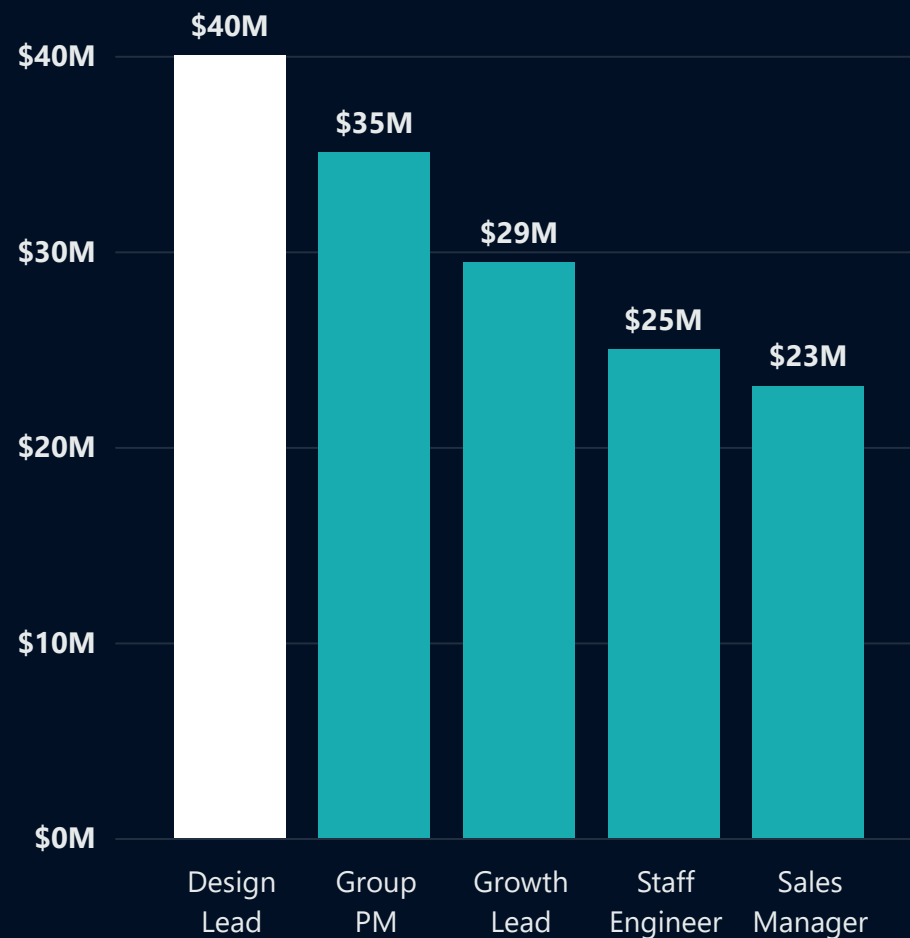


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Atlas TechWorks - Design Lead Positions Carry the Highest Attrition Costs

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Top 5 Job Roles by Annual Attrition Cost



INSIGHTS

Design Leads account for approximately \$40M in annual attrition costs, making them the most expensive role to replace.

Group Product Managers and Growth Leads follow closely behind. Since these positions are highly specialized, losing employees in these roles creates substantial financial and operational impacts.

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Atlas TechWorks - Engineering, Product, and Sales Departments Carry the Largest Attrition Costs

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Replacement Cost Breakdown

	Engineering	\$63.5M	31.30%
	Product	\$62.4M	26.04%
	Sales	\$57.4M	28.46%
	Marketing	\$56.0M	31.61%
	Design	\$48.9M	27.43%
	Customer Success	\$25.6M	35.29%
	Operations	\$20.6M	30.63%
	Finance	\$15.2M	34.45%
	HR	\$13.5M	29.01%

INSIGHTS

Engineering incurs the highest annual attrition cost at \$63.5M, followed closely by Product and Sales. Despite Customer Success recording the highest attrition rate, its total cost impact is lower due to differences in salary structures and replacement expenses.

These findings indicate that cost reduction efforts should prioritize departments with the largest financial exposure.

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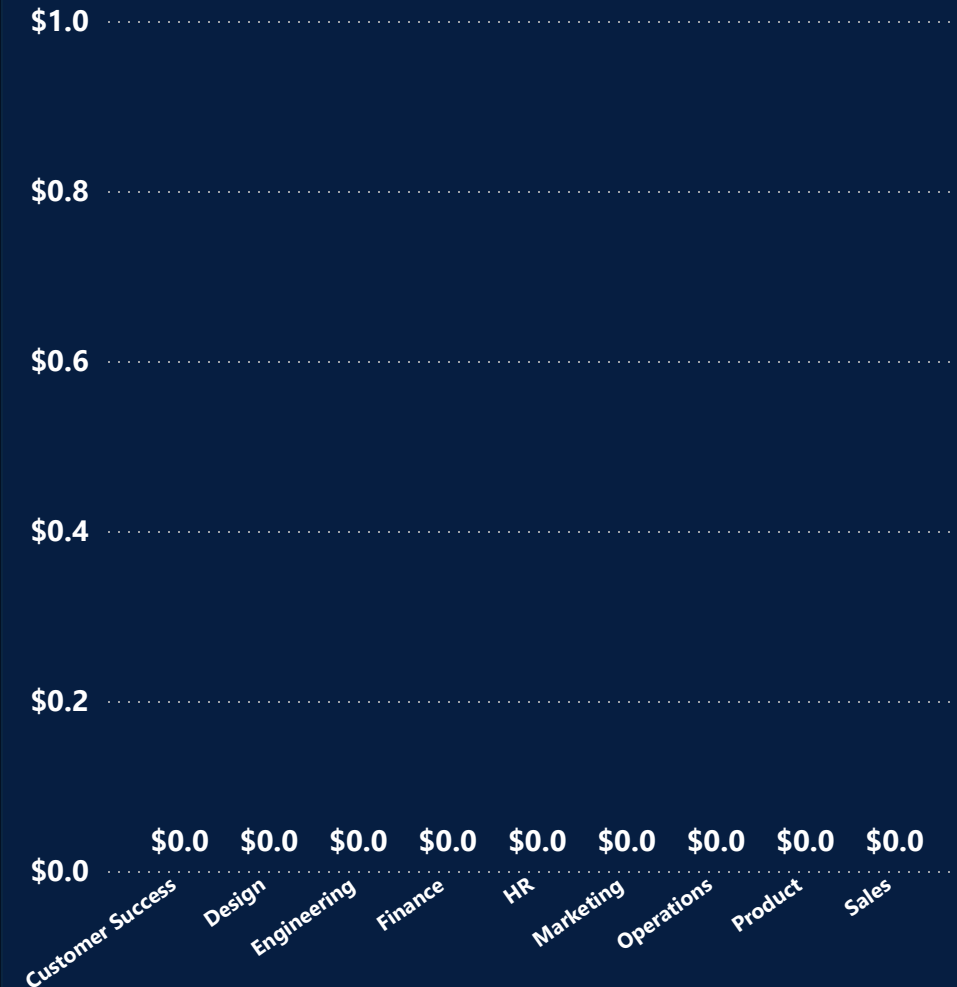


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Atlas TechWorks - Retention Interventions Create Potential Savings Across Departments

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Simulator Total Savings by DepartmentName



INSIGHTS:

The simulation dashboard estimates the financial benefits that Atlas TechWorks could realize by reducing attrition through targeted interventions. Depending on the selected scenario and fix intensity, each department displays a different level of potential savings, reflecting the varying impact of turnover across the organization.

Departments with larger workforces, higher attrition rates, or more expensive replacement costs are expected to generate the greatest savings when retention initiatives are implemented successfully. In contrast, departments with lower turnover or smaller employee populations contribute comparatively smaller savings opportunities.

This analysis demonstrates that attrition reduction is not only a workforce challenge but also a major financial opportunity. By focusing interventions on high-impact groups such as employees affected by excessive overtime, low job satisfaction, poor work-life balance, or promotion stagnation Atlas TechWorks can significantly reduce turnover costs and preserve valuable talent.

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HEADCOUNT



1,500

ATTRITION RATE



30.13%

TOTAL ATTRITION



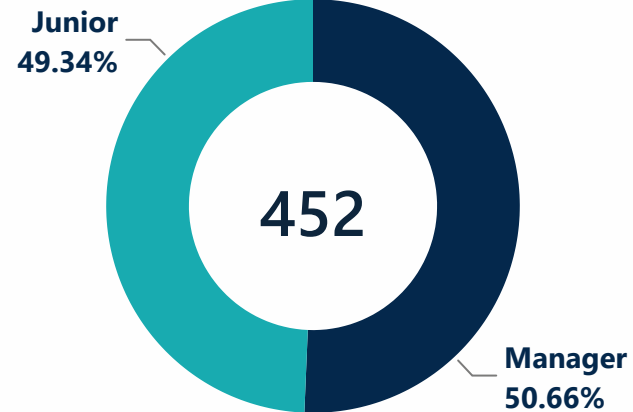
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Total Attrition Cost

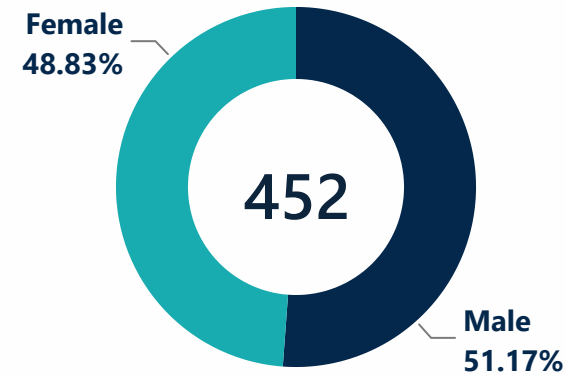


\$363.2M

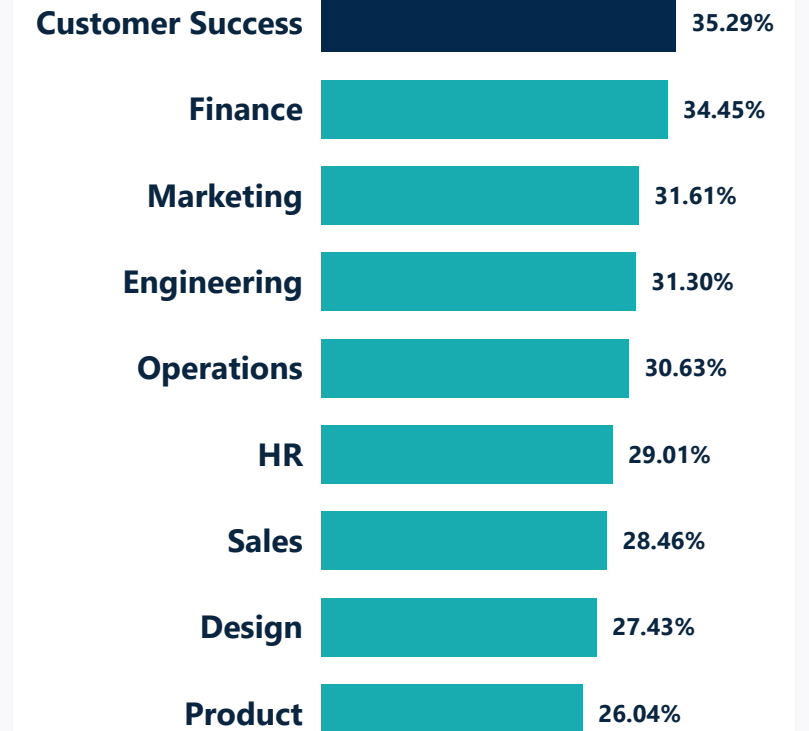
Attrited Employees by JobLevel



Attrition Rate by Gender



Attrition Rate by Department Name





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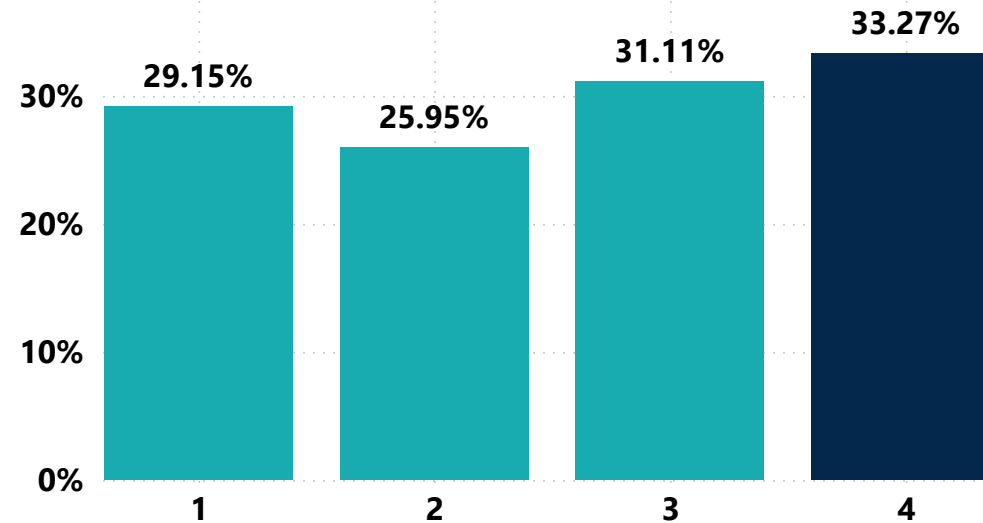
Cost Analysis



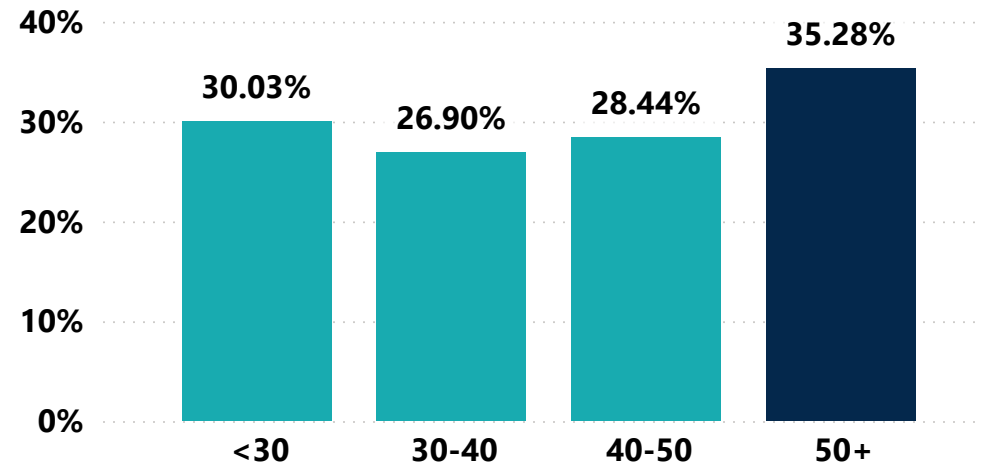
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Attrition Rate by Tenure Band



Attrition Rate by Age Band



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TOTAL ATTRITION



452

Total Attrition Cost



\$363.2M

Top Ten Job RoleName by Attrition Rate

JobRoleName	DepartmentName	Attrition Rate	Attrited Employees
Support Engineer	Customer Success	38.98%	23
Software Engineer	Engineering	38.71%	24
Sales Manager	Sales	36.62%	26
Product Manager	Product	36.54%	19
Marketing Specialist	Marketing	36.51%	23
Financial Analyst	Finance	36.36%	20
Design Lead	Design	35.82%	24
Account Executive	Sales	33.33%	19
Controller	Finance	32.81%	21
Senior Engineer	Engineering	32.08%	17



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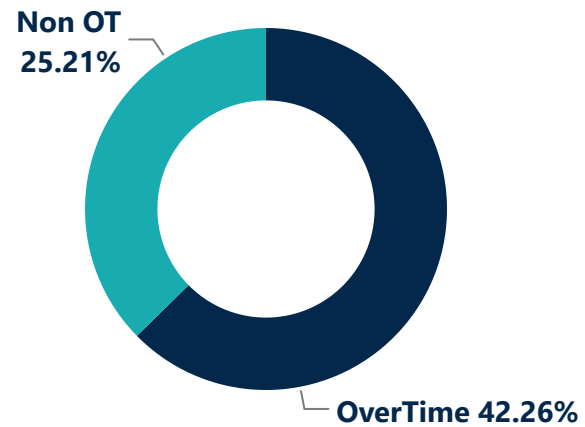
Why are they Leaving?



OverTime Impact

Employees Working overtime have a significant higher attrition rate

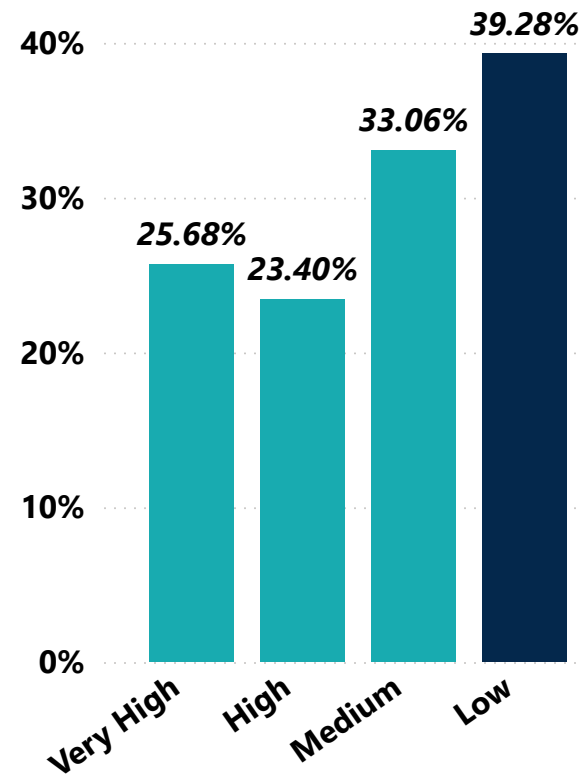
Attrition Rate by Overtime



Job Satisfaction Level

Employees with the lowest job satisfaction level are most likely to leave

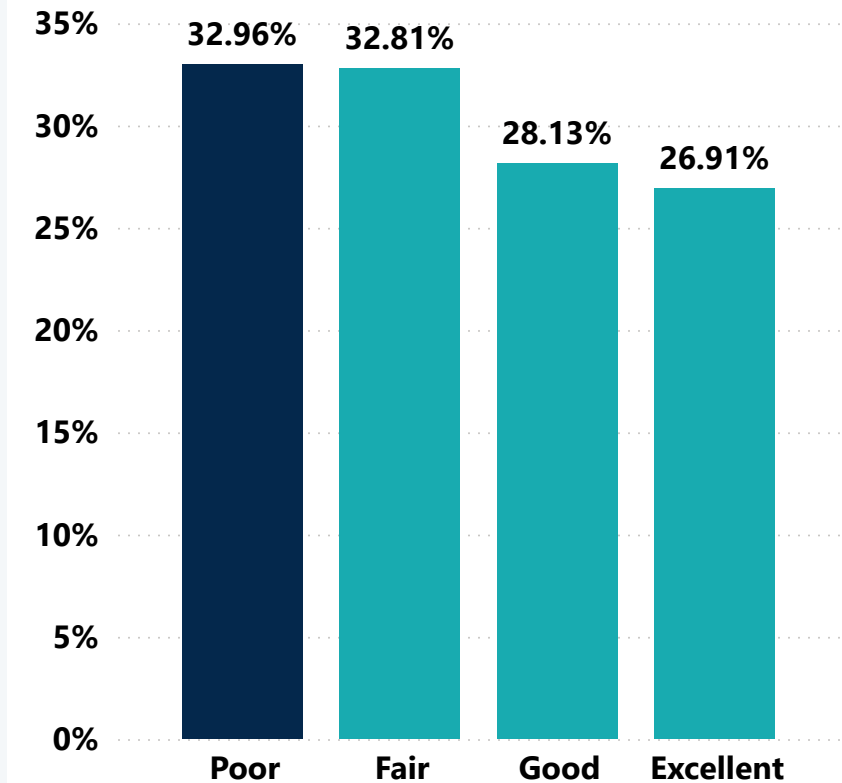
Attrition Rate by Job Satisfaction



Work-Life Balance vs Attrition

Poor work-life balance is strongly associated with higher attrition

Attrition Rate by WorkLifeBalance



Attrition Rate by Distance Band

Very Near

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Cost of Attrition Analysis

Financial Impact of employee attrition on Atlas TechWorks

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Total Annual Attrition Cost

\$363.2M



Replacement Cost Breakdown



Engineering

\$63.5M

31.30%



Product

\$62.4M

26.04%



Sales

\$57.4M

28.46%



Marketing

\$56.0M

31.61%



Design

\$48.9M

27.43%



Customer Success

\$25.6M

35.29%



Operations

\$20.6M

30.63%



Finance

\$15.2M

34.45%



HR

\$13.5M

29.01%

Annual Attrition Cost by Top 5 JobRoleName



Avg Attrition Cost per E...

\$803.6K



Attrited Employees

452



% Payroll Lost

30.15%



Total Annual Payroll

192M





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Cost of Attrition Analysis

Financial Impact of employee attrition on Atlas TechWorks

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SCENARIO

Eliminate High Impact Groups

POTENTIAL ANNUAL SAVINGS

\$0.0

EMPLOYEE RETAINED

0

PROJECTED ATTRITED
EMPLOYEES

452

Adjust Intervention Lever

0%



Scenario Name

Address
Stagnation

Improve Job
Satisfaction

Elimate
Overtime

Improve
Worklife
Balance

Simulator Total Savings by DepartmentName

\$1.0

\$0.5

\$0.0 \$0.0 \$0.0 \$0.0 \$0.0 \$0.0 \$0.0 \$0.0 \$0.0 \$0.0

Customer Success Design Engineering Finance HR Marketing Operations Product Sales

Projected Attrition Rate

30.13%

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INSIGHT

The workforce analysis reveals that Atlas TechWorks is experiencing a relatively high attrition rate, with 452 employees leaving and generating an estimated annual replacement cost of \$363.2M. Attrition affects both junior and managerial employees almost equally, indicating that retention challenges are present across multiple organizational levels.

Customer Success records the highest attrition rate, while Engineering, Product, and Sales account for the greatest financial impact. Support Engineers and Software Engineers emerge as some of the most vulnerable roles, highlighting risks within critical technical functions.

Further analysis shows that overtime, low job satisfaction, Far distance from Home and poor work-life balance are strongly associated with higher attrition. Employees with longer tenure and those aged 50 and above also exhibit elevated turnover rates, suggesting that career stagnation and changing career priorities may be contributing factors.

Overall, the findings indicate that employee turnover is driven more by workplace conditions and career development challenges than by demographic differences, presenting clear opportunities for targeted retention strategies.

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RECOMMENDATION

To reduce employee turnover and protect talent investments, Atlas TechWorks should prioritize initiatives aimed at improving employee experience and career growth. Excessive overtime should be managed carefully to minimize burnout, while programs that enhance job satisfaction and work-life balance should be strengthened.

Departments with high turnover rates, particularly Customer Success, should receive targeted retention support, while high-cost departments such as Engineering, Product, and Sales should be prioritized to minimize financial losses. Special attention should also be given to critical roles like Support Engineers and Software Engineers, where turnover poses operational risks.

Promotion pathways and career development opportunities should be reviewed to address stagnation among long-serving employees. Leveraging the attrition simulator to evaluate intervention scenarios will help management focus resources on areas with the greatest potential return. Continuous monitoring through the dashboard will enable Atlas TechWorks to build a more stable workforce, reduce replacement costs, and sustain long-term business performance.

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Atlas TechWorks - Closing Remark

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